04/25/2022

Monday, April 25, 2022 9:01 AM

M168_Notes_5



## ***Section 5.1 Compound Interest***

***Simple interest*** – interest computed as a percentage of the principal

1% interest

$100 → $101 After a yr.

Simple interest earned: $I = Prt$

$I$=interest earned
$P$=principal
$r$=interest rate (decimal form)
$t$=time period (in years)

Future value: $F = P + Prt = P(1 + rt)$

*Example*
If a bank loans $525 to an individual for 3½ years at 7.25% simple interest, what will be the amount repaid on the loan?

$P = 525$
$r = 0.0725$
$t = 3.5$

$F = 525(1 + 0.0725 \times 3.5)$
$= 658.22$

*Example*
If a $1000 deposit grows in value to $1024 after 9 months, what is the simple interest rate that is earned?

$P = 1000$
$r =$ unknown
$t = 0.75$

$1000(1 + r \times 0.75) = 1024$

$1000 + r \times 0.75 \times 1000 = 1024$

$r = \frac{24}{0.75 \times 1000} = \frac{24}{750} = 3.2$

$r = 3.2\%$

*Example*

How much should be place in an account that pays simple interest of 4% so that the value of the account after 18 months is $3000?

$P =$ Unknown
$r = 0.04$
$t = 1.5$

$$P(1 + 0.04 \times 1.5) = 3000$$
$$\Rightarrow P = \frac{3000}{1 + 0.04 \times 1.5} = \boxed{2830.189}$$

*Example*

You have an account with $500 that pays 5% simple interest. How long until your account doubles in value?

$P = 500$
$r = 0.05$
$t =$ Unknown

$$P(1 + 0.05\,t) = 2P$$
$$\Rightarrow 0.05\,t = 1$$
$$\Rightarrow t = \frac{100}{5} = \boxed{20} \leftarrow \text{20 years.}$$

When interest is paid periodically and the interest earns interest, we have compound interest.

*Example*

You deposit $100 into an account that pays 10% annual interest that is compounded annually. How much is in the account after 4 years?

$$F_1 = 110$$
$$F_2 = 110(1 + 0.1) = 121$$
$$F_3 = 121(1 + 0.1) = 133.1$$
$$F_4 = 133.1(1 + 0.1) = 146.41$$

$m=1$

$$F = P\left(1+\frac{r}{m}\right)^{mt}$$

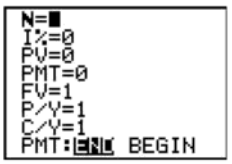
where $m$ is the number of compounding periods per year and $r$ is the annual interest rate as a decimal.

*Example*

You deposit \$300 into an account that pays 6% annual interest compounded monthly. How much is in the account after 3 months?

| If interest is compounded | Then $m$ is |
|---|---|
| Annually | 1 |
| Semiannually | 2 |
| Quarterly | 4 |
| Monthly | 12 |
| Weekly | 52 |
| Daily | 365 |

ON THE CALCULATOR, TVM Solver:




N is

I% is

PV is

PMT is

FV is

P/Y and C/Y is

set PMT: END (make payments at the end of the cycle)

*Example*
You deposit $100 into an account that pays 10% annual interest that is compounded annually. How much is in the account after 4 years?

*Example*
You deposit $300 into an account that pays 6% annual interest compounded monthly. How much is in the account after 3 months?

*Example*

You deposit $1000 into an account that pays 10% annual interest. How much is in the account after 10 years if

(a) the account earns simple interest?

(b) the account earns interest compounded annually?

(c) the account earns interest compounded semiannually?

(d) the account earns interest compounded quarterly?

(e) the account earns interest compounded monthly?

(f) the account earns interest compounded weekly?

(g) the account earns interest compounded daily?

Effective interest rate $r_{eff}$ can be found with a FINANCE command

Eff( I%, m)

*Example*

You find a bank that pays 8.15% annual interest compounded quarterly and another that pays 8.1% annual interest compounded daily. Which is a better deal?

If interest is compounded continuously, then the future value of an investment of P dollars at an annual interest of $r$ (expressed as a decimal) for $t$ years is $F = Pe^{rt}$.

*Example*

You deposit $1000 into an account that pays 10% annual interest. How much is in the account after 10 years if the interest is compounded continuously?

*Example*

You wish to have $60,000 available for your child's college expenses. You find an account that pays 7% annual interest compounded monthly.

How much needs to be deposited now to reach this goal if you want to have the money available in 18 years?

How much interest was earned?

What if you wanted the money available in 8 years?

How much interest was earned?

*Example*
You deposit $4000 in an account that earns 5% annual interest compounded weekly. How long does it take for the account to double in value?

*Example*
You deposited $200 in a savings account 10 years ago. It is now worth $350. You know the account paid annual interest that was compounded annually, but you don't remember the interest rate promised. What was the annual interest rate for the account?

*Example*
Your retirement account was worth $16,000 on January 1 and worth $15,000 two years later. Find the annual interest rate.

*Example*

To save for retirement, you deposit $2000 every year in an account that earns 6% annual interest compounded annually.

(a) How much will you have after 10 years? How much interest did you earn?

(b) How much will you have after 25 years? How much interest did you earn?

(c) How much will you have after 40 years? How much interest did you earn?

An ***annuity*** is an account to which regular payments are made. An annuity that is ***certain and simple*** has the following properties:

1. The payments are made at fixed time intervals
2. The periodic payments are of equal size
3. The payments are made at the end of the interval
4. The interest is paid at the end of the interval

Many loans and savings plans are certain and simple annuities

Any account that is established to meet a future need is called a ***sinking fund***.

*Example*
You wish to have $60,000 available for your child's college expenses. You find an account that pays 7% annual interest compounded monthly. How much needs to be deposited every month to reach this goal if you want to have the money available in 18 years? Assume that the monthly payments are of equal size. How much interest does the account earn?

*Example*
You start a savings account with the goal of saving $5000 in 4 years by making regular quarterly payments to an account that earns 8% annual interest compounded quarterly.

(a) How large are the quarterly payments?

(b) How much interest is earned in all?

(c) How much interest is earned in the 3rd quarter of the 2nd year?

*Example*
A cello costs $574 to buy or $40.58 per month on a 2 year rent-to-own plan. What is the interest rate charged?

*Example*
You owe $6000 on a credit card that charges 21% annual interest compounded monthly on the unpaid balance.

(a) If you make monthly payments of $120, how long will it take to pay it off?

(b) How much interest is paid in all?

(c) How much interest do you own after 5 years of making payments?

*Example*
You want a pair of diamond earrings that cost $800. You decide to stop buying daily lattes and instead save the $5 every day in an account that pays 6% annual interest compounded daily. How long until you have the money for the earring?

*Example*

When you retire you would like to have $30,000 per year. Assuming that you will need to have 20 payments starting right now, how much do you need to deposit into an account that pays 5% annual interest that is compounded annually?

*Example*

You wish to purchase a house that costs $130,000. You make a down payment of $20,000 and finance the remainder for 30 years at 5.1% annual interest compounded monthly on the unpaid balance.

(a) How large are the monthly payments?

(b) How much interest is paid in all?

(c) How much of the first payment is interest?

(d) How much do you still owe the bank after your first payment?

(e) What is your equity after your first payment?

(f) What is your equity after your second payment?

(g) What is your equity after your $12^{th}$ payment?

*Example*

You borrow $3000 at 12% annual interest compounded quarterly for 2 years. Show in a table how much of each payment is interest, how much is paid towards the principal, how much of the loan is paid off and how much is still owed each quarter.

| end of period | # PMT left | PMT ($) | interest ($) | towards principal | outstanding principal | equity |
|---|---|---|---|---|---|---|
| 0 | | | | | | |
| 1 | | | | | | |
| 2 | | | | | | |
| 3 | | | | | | |
| 4 | | | | | | |
| 5 | | | | | | |
| 6 | | | | | | |
| 7 | | | | | | |
| 8 | | | | | | |

*Example*

For the house purchased earlier, fill in the following lines of an amortization table.

| end of period | # PMT left | PMT ($) | interest ($) | towards principal | outstanding principal | equity |
|---|---|---|---|---|---|---|
| 0 | | | | | | |
| 1 | | | | | | |
| 2 | | | | | | |
| | | | | | | |
| 12 | | | | | | |
| | | | | | | |
| 60 | | | | | | |
| | | | | | | |
| 120 | | | | | | |
| | | | | | | |
| 180 | | | | | | |
| | | | | | | |
| 240 | | | | | | |
| | | | | | | |
| 300 | | | | | | |